

TA Composite

2026-08-10

Alpha TA · ASH · BTC-DC · Cup & Handle · PCR Short | full tested period 2017-2026 (8.8y) | causal replay, 8.5bp/side + real funding

CAGR
+26.5%
annualised

Max Drawdown
-13.0%
worst peak-to-trough

Sharpe
1.67
annualised

MAR
2.04
CAGR / MaxDD

Total Return
+700%
8.8 years

Sortino
2.71
downside-adjusted

Volatility
14.7%
annualised

Longest DD
221d
median spell 6d

Equity curve — portfolio vs sleeves (log scale, 1x)



Portfolio drawdown



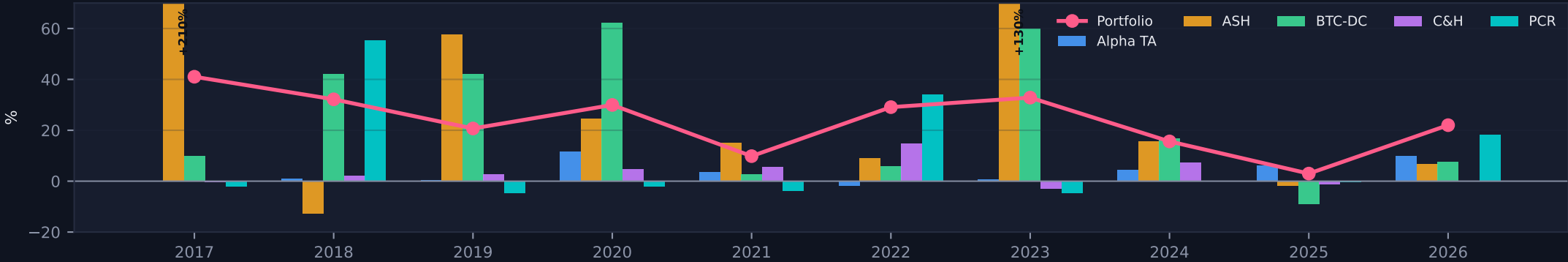
Sleeve	Notional wt	Risk share	CAGR	MaxDD	Sharpe	MAR	Longest DD
Alpha TA	86%	14%	+4.0%	-6.4%	0.76	0.62	773d
ASH	29%	54%	+39.1%	-38.5%	1.11	1.02	560d
BTC-DC	14%	13%	+24.9%	-17.5%	1.13	1.42	439d
Cup & Handle	57%	8%	+3.6%	-11.0%	0.57	0.33	550d
PCR Short	50%	10%	+8.5%	-12.1%	1.05	0.70	1228d
XS MeanRev (retired)	—	—	+4.2%	-40.1%	0.31	0.10	1943d
BTC buy & hold	—	—	+35.9%	-82.0%	0.80	0.44	1095d
PORTFOLIO (Option B)	186%	100%	+26.5%	-13.0%	1.67	2.04	221d

Notional weight is exposure; risk share is the contribution to portfolio volatility. A low-drawdown sleeve can carry a large notional weight cheaply — ASH is the largest RISK position despite the smallest weight.

Yearly & Monthly Breakdown

per-year returns by sleeve, portfolio monthly grid, and return distribution

Annual return by sleeve (1x, unweighted) — bars clipped at +70%



Portfolio monthly returns (%)

2017	-	-	-	-	-	-	-	-	+0.6	+9.9	+9.8	+16.2	+41.0
2018	+7.6	-0.1	+6.3	+3.7	+3.9	+2.7	+3.4	+6.6	-6.9	-2.3	+11.8	-6.6	+32.2
2019	+0.8	+1.8	+1.0	+3.9	+13.5	+3.1	+0.1	-1.8	-1.4	-0.5	-0.2	-0.6	+20.7
2020	+5.6	+0.1	+1.1	+1.4	+1.1	-5.0	+5.4	-1.4	+0.1	+14.5	+3.0	+1.7	+30.0
2021	+2.3	+1.1	-0.7	-0.4	-3.0	+0.8	-0.4	+5.6	-4.3	+6.0	+1.2	+1.6	+9.8
2022	+5.5	+3.0	+3.5	+0.4	+8.9	+9.8	-2.3	-3.2	-1.2	-3.1	+3.8	+1.6	+29.1
2023	+15.1	-0.7	+2.9	+0.7	-1.0	+5.4	-1.0	-3.5	+0.3	+8.7	-0.4	+3.6	+32.8
2024	-0.4	+5.1	-0.0	-0.1	+0.1	-1.5	+2.3	+2.0	+0.7	-1.1	+6.1	+1.6	+15.6
2025	-4.8	+5.9	-0.3	-0.2	+7.2	+0.3	-1.6	-1.7	-1.8	-0.2	+2.0	-1.2	+3.0
2026	+8.5	+4.1	-1.5	-2.9	+1.9	+11.6	-0.1	-0.5	-	-	-	-	+22.0
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Year

Daily return distribution (active days)



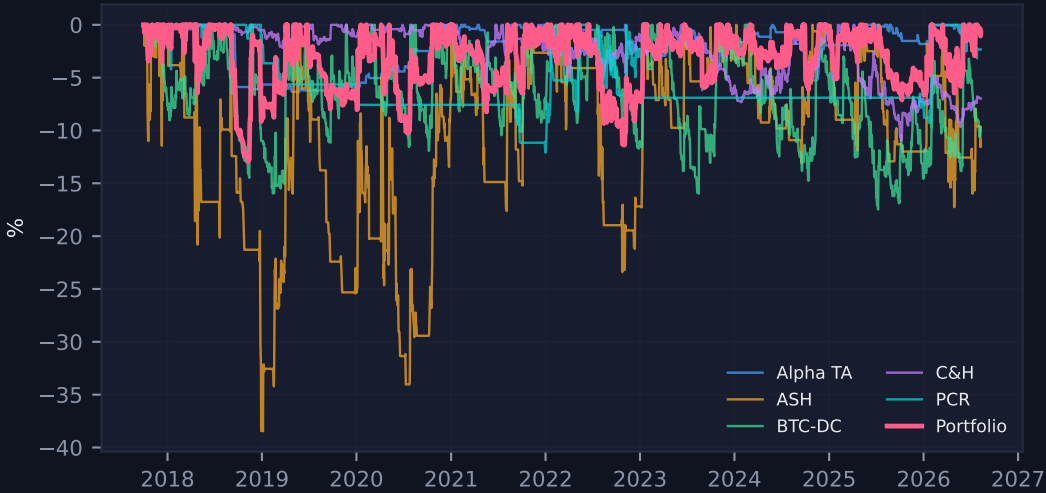
Trade Statistics, Risk & Correlation

per-trade behaviour of the signal sleeves, drawdown profile, and sleeve correlations

Per-trade statistics (full history, 1x, net of costs)

Sleeve	Trades	/yr	Win rate	Profit factor	Avg trade	Avg win	Avg loss	Best	Worst	Avg hold
ASH	379	82	41.7%	2.08	+1.06%	+4.91%	-1.68%	+32.1%	-10.8%	2.1d
BTC-DC	373	81	37.0%	1.55	+1.13%	+8.57%	-3.24%	+58.8%	-20.3%	4.6d

Drawdown — portfolio vs sleeves



Sleeve correlation (daily, full period)

ATA	1.00	0.12	0.14	0.02	0.03
ASH	0.12	1.00	0.42	-0.05	-0.05
DC	0.14	0.42	1.00	0.17	0.26
C&H	0.02	-0.05	0.17	1.00	0.26
PCR	0.03	-0.05	0.26	0.26	1.00
	ATA	ASH	DC	C&H	PCR

Risk summary — portfolio

Risk metric	Value	Risk metric	Value
Annualised volatility	14.7%	Longest drawdown	221 days
Best day	+8.05%	Median drawdown spell	6 days
Worst day	-5.24%	Sortino ratio	2.71
Positive days (of active)	48.4%	Design DD budget	-8.5% (ceiling -10%)

Results by window

Window	CAGR	MaxDD	Sharpe	Total
Full tested period 2017-2026 (8.8y)	+26.5%	-13.0%	1.67	+700%
Out-of-sample only 2022-2026 (4.6y)	+21.9%	-11.4%	1.56	+149%
BTC buy & hold 2017-2026	+35.9%	-82.0%	0.80	+1416%

The headline figures on page 1 use the full tested period. The 2022-2026 row is the stricter out-of-sample cut: ASH and BTC-DC parameters were chosen knowing the earlier history, so pre-2022 is not strictly out-of-sample. Both are shown so the reader can judge; neither was selected for how it looks.

Methodology, Corrections & Known Limitations

what this backtest does, what was fixed, and what is still uncertain

METHOD

Bar-level mark-to-market on 4h bars (daily for the Alpha TA OPA sleeve), compounded daily.

Every position array is shifted one bar before multiplying by returns, so a signal generated at a bar close earns its first return in the NEXT bar — matching how the live bot executes.

Costs: 8.5bp per side (Hyperliquid 3.5bp taker + 5bp assumed slippage) on every notional change.

Funding: real Binance 8h funding series, signed — longs pay when the rate is positive, shorts receive.

Window: headline figures use the FULL tested period, 2017-09 to 2026-08 (8.8 years).

The stricter 2022-2026 out-of-sample cut is shown alongside on page 3. ASH and BTC-DC are frozen specs, so both windows use identical rules — only the scoring span differs.

Sleeves were scaled so the combined drawdown lands at -8.5% ON THE 2022-2026 WINDOW. Over the full 8.8 years the same weights draw down -14.2% — a concrete demonstration that a single-path backtest drawdown understates what a full cycle delivers.

CORRECTIONS MADE (2026-08-04)

Cup & Handle look-ahead: the replay let the entry bar earn its own return though entry happens AT that close. Fixing it moved C&H OOS Sharpe 1.17 -> 0.52. Prior portfolio weights were built on the inflated figure and were re-fitted from scratch.

USDT.D regime gate REMOVED from ASH: USDT dominance is mechanically the inverse of the same-day crypto return (measured same-bar correlation -0.96 on real 4h data). A daily value stamped 00:00 embeds the whole day, so gating that day's bars leaked direction. Lagging it one day collapsed Sharpe 1.15 -> 0.61. Replaced with the SMA200(4h) gate from ASH's original frozen spec.

XS Mean Reversion RETIRED: modelled rather than assumed, it scored Sharpe 0.18 with a -34.5% drawdown and pushed portfolio DD from -8.45% to -10.17% for +0.3pp of return.

Alpha TA RESTORED: dropped earlier against the two inflated sleeves above. On clean numbers it is weak alone (Sharpe 0.72) but nearly uncorrelated (0.15 / 0.16 / 0.01) and rises as the trend sleeves decay — positive in 2025 and strongest in 2026.

AUDIT PASSED

Fear & Greed lag test: Sharpe 1.08 -> 1.12 under a one-day lag (a look-ahead signal collapses).

Same-day price embedding: F&G correlates +0.02 same-day and +0.57 prior-day — the correct signature.

Full-sample constants: the F&G normaliser is a pure scalar; Sharpe is unchanged across its range.

BTC-DC engine verified identical to the source implementation across all 19,328 bars.

KNOWN LIMITATIONS

Survivorship: the Cup & Handle universe is TODAY'S 30-coin list. Coins that existed in 2022 and later delisted are absent entirely. This biases C&H upward and is NOT corrected — it needs a point-in-time universe we do not have. C&H carries 57% weight, so this is the largest open risk.

Re-fitting: this portfolio was re-fitted several times against the same 2022-2026 window during a single research session. Each pass consumes some of the out-of-sample validity being relied upon.

ASH decay: ASH scores Sharpe ~1.9 in 2022-23 and roughly 0.0-0.6 from 2024 onward under every gate tested. It still earns its weight over the full window but is not currently performing.

Fills are modelled at bar close. Real slippage on Hyperliquid is assumed, not measured.

No live track record yet. Every figure here is simulation.

Drawdown budget vs reality: the -8.5% design target was set on 2022-2026. Judged over the full history the deployed weights run -14.2%, i.e. roughly 1.7x the intended risk. The weights were knowingly left unchanged; holding -8.5% across a full cycle would require scaling to 0.60x, which would cut CAGR from +21.5% to +12.7% at the same Sharpe.